

Research Paper Summarizer using n8n, arXiv & Google Workspace

In the digital era, researchers face the challenge of navigating through an overwhelming volume of scientific papers. Finding, reading, and summarizing relevant research manually can be time-consuming and inefficient.

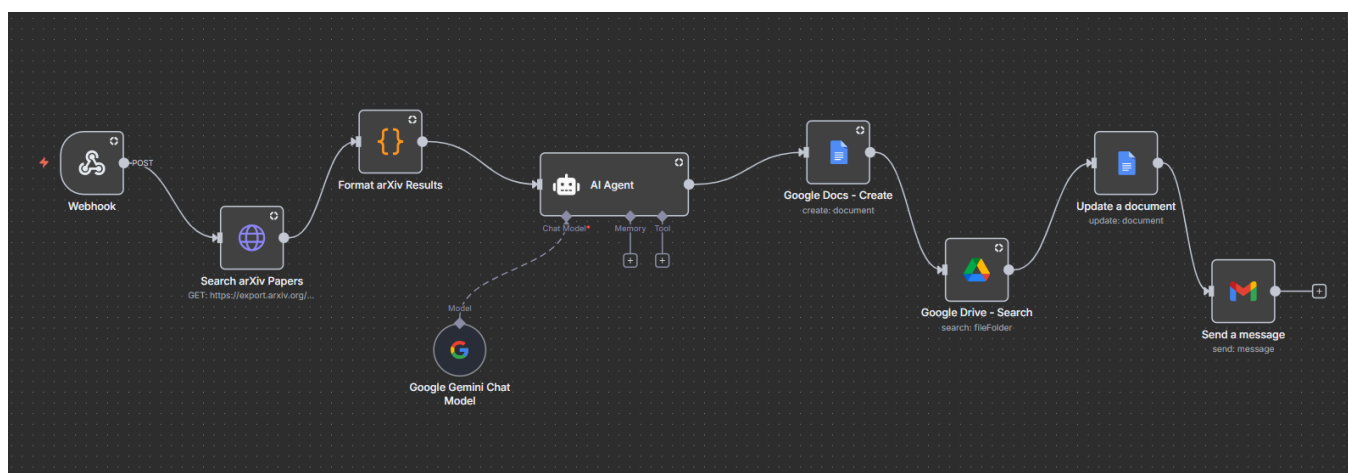
This project addresses that challenge by creating an **automated research assistant** that retrieves scholarly articles from **arXiv**, summarizes them using **Gemini AI**, and delivers the results directly to the user's **Google Drive** and **Gmail** — all orchestrated using **n8n**, a no-code automation platform.

The solution demonstrates the seamless integration of multiple APIs and AI capabilities to create a smart research automation workflow.

Objectives

- Automate the retrieval of research papers based on a user-defined topic.
- Generate concise, human-like summaries using an AI model.
- Store and manage summarized results in Google Docs and Drive.
- Automatically share the document link via Gmail.
- Provide a scalable and low-code workflow solution using n8n.

Workflow Overview



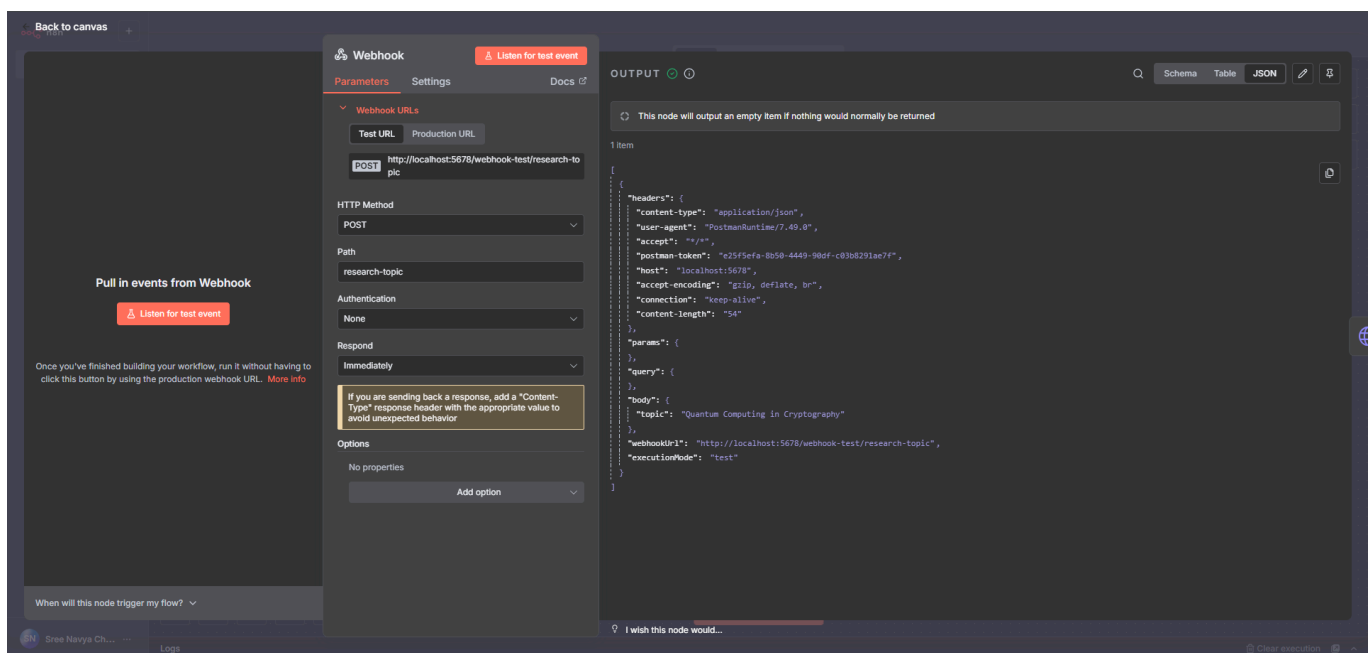
The **Research Paper Summarizer** automates five major tasks:

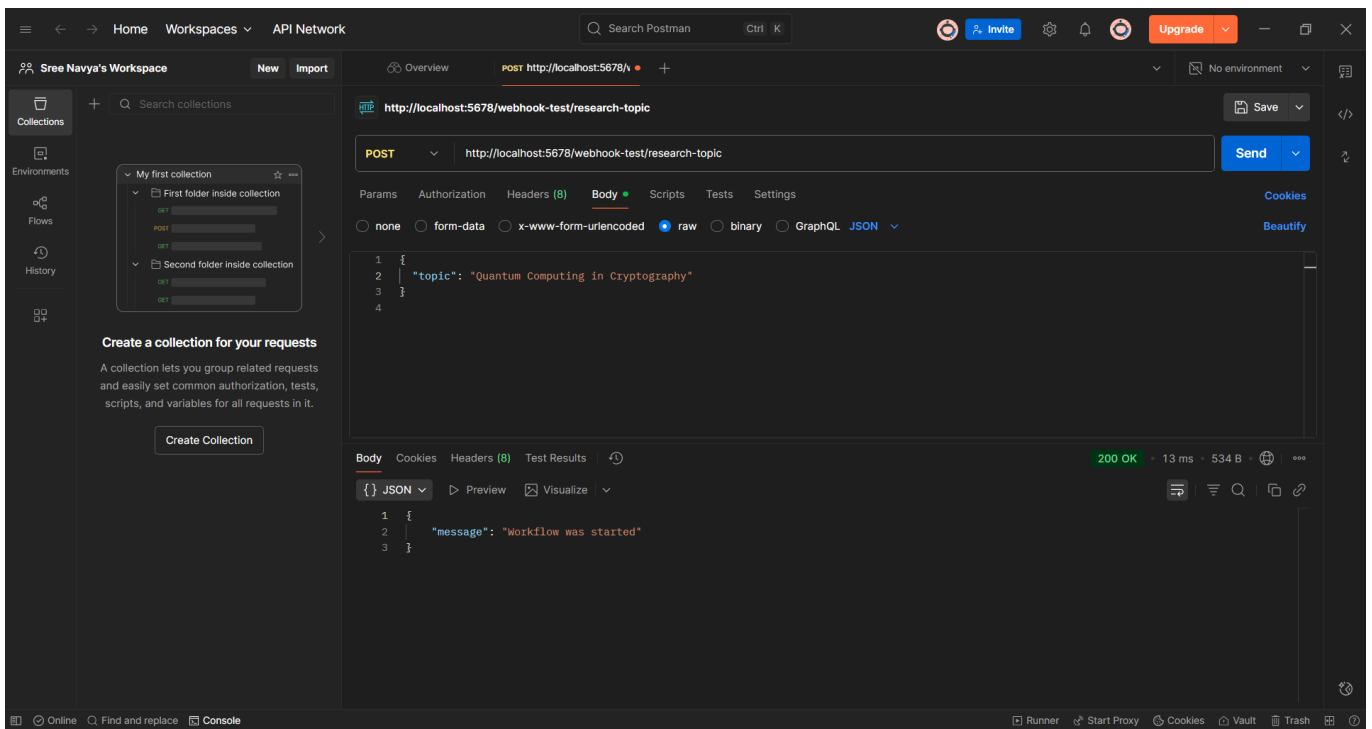
1. **Trigger** – Receives a research topic input from Postman via n8n Webhook.
2. **Search** – Queries the **arXiv API** to fetch the top 5 research papers related to the topic.
3. **Summarize** – Uses **Gemini AI** to summarize all retrieved papers into a unified, readable summary.
4. **Document Creation** – Creates and updates a **Google Doc** with the summarized output.
5. **Email Delivery** – Sends the document link to the user’s Gmail inbox automatically.

1 Webhook Node — “Receive Research Topic”

Purpose:

Acts as the entry point of the entire workflow. This node listens for incoming HTTP POST requests (like from Postman) and starts the automation when data is received.

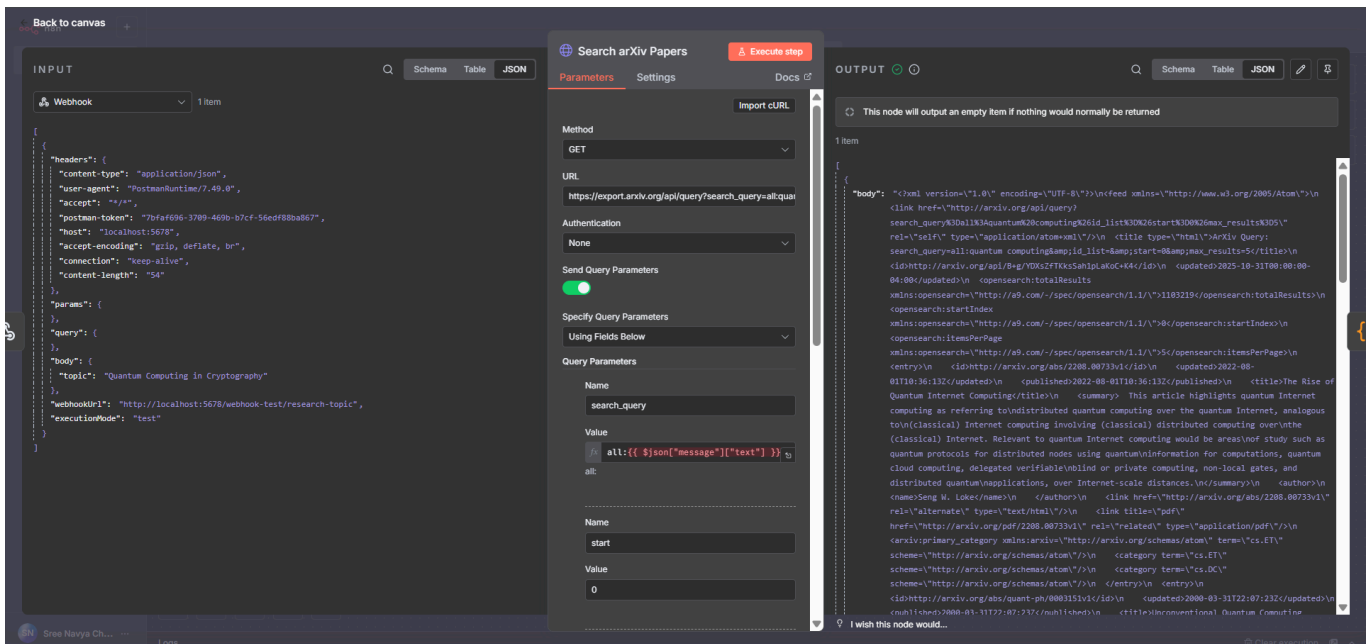




2 HTTP Request Node — “Search arXiv for Papers”

Purpose:

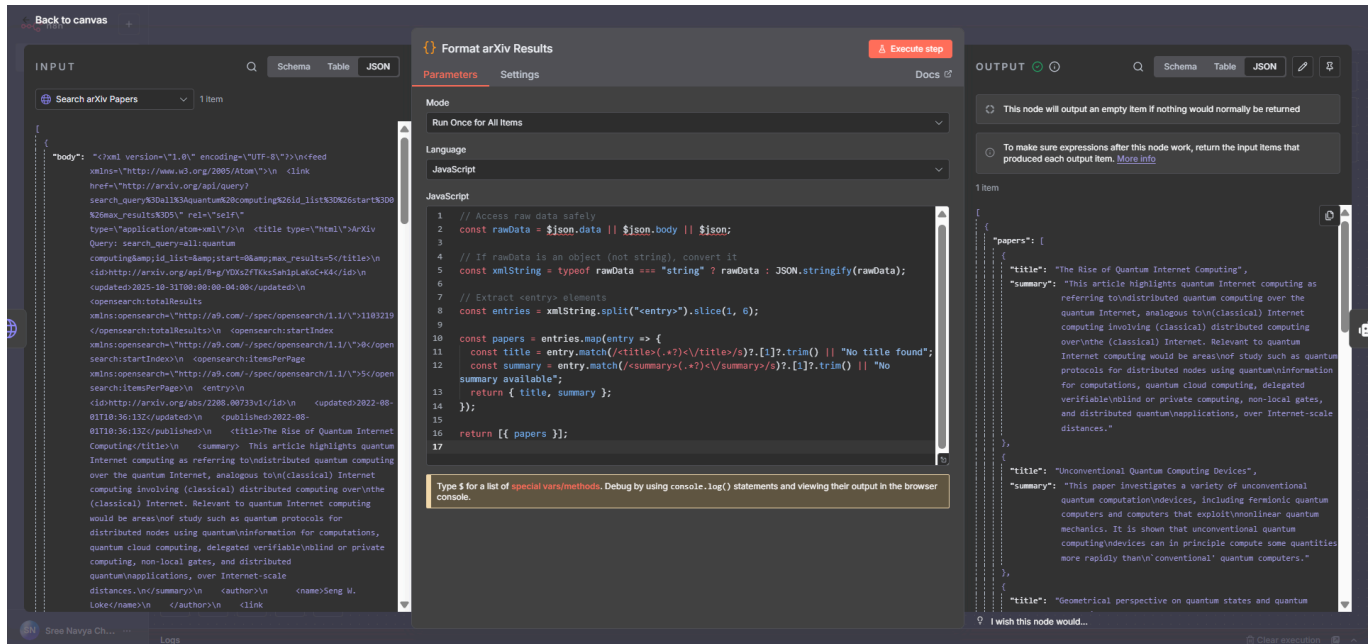
Queries the **arXiv API** using the received research topic to fetch scholarly papers related to it.



3 Code Node — “Format arXiv Data”

Purpose:

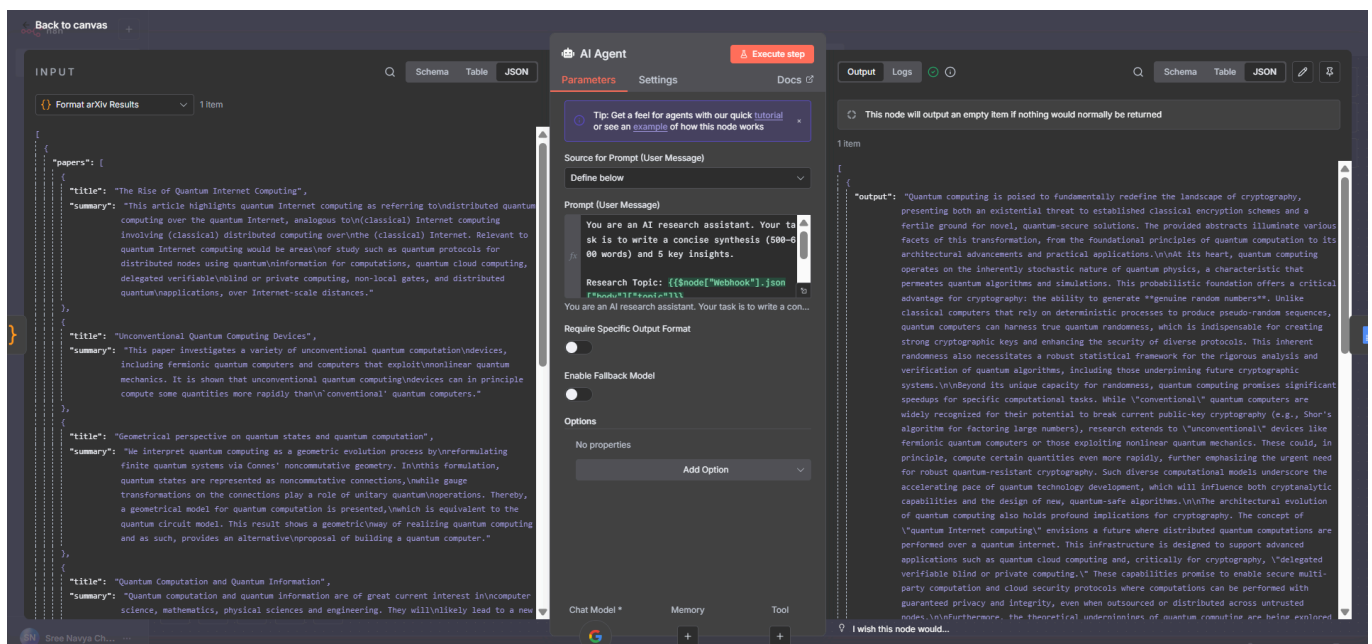
Parses the XML response received from arXiv and formats it into readable text for the AI summarization step.



4 AI Agent Node — “Summarize with Gemini”

Purpose:

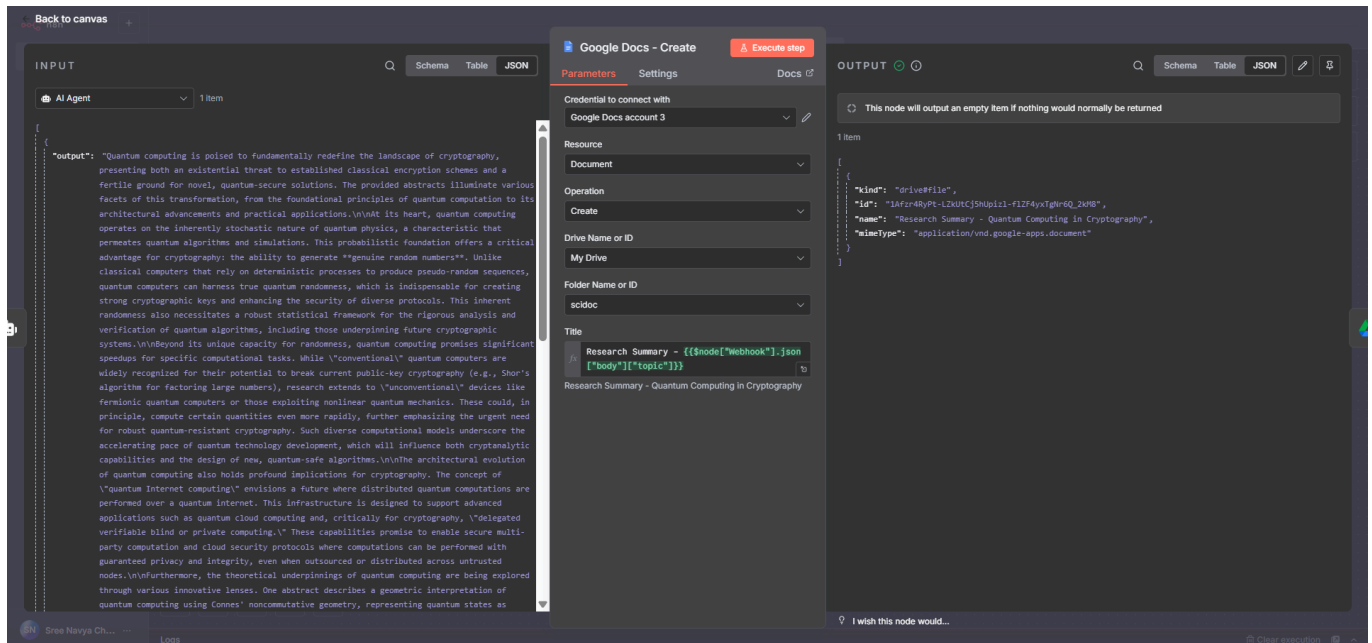
Uses **Gemini AI (Google Generative AI API)** to read all the extracted paper summaries and generate a concise combined summary.



5 Google Docs Node — “Create Document”

Purpose:

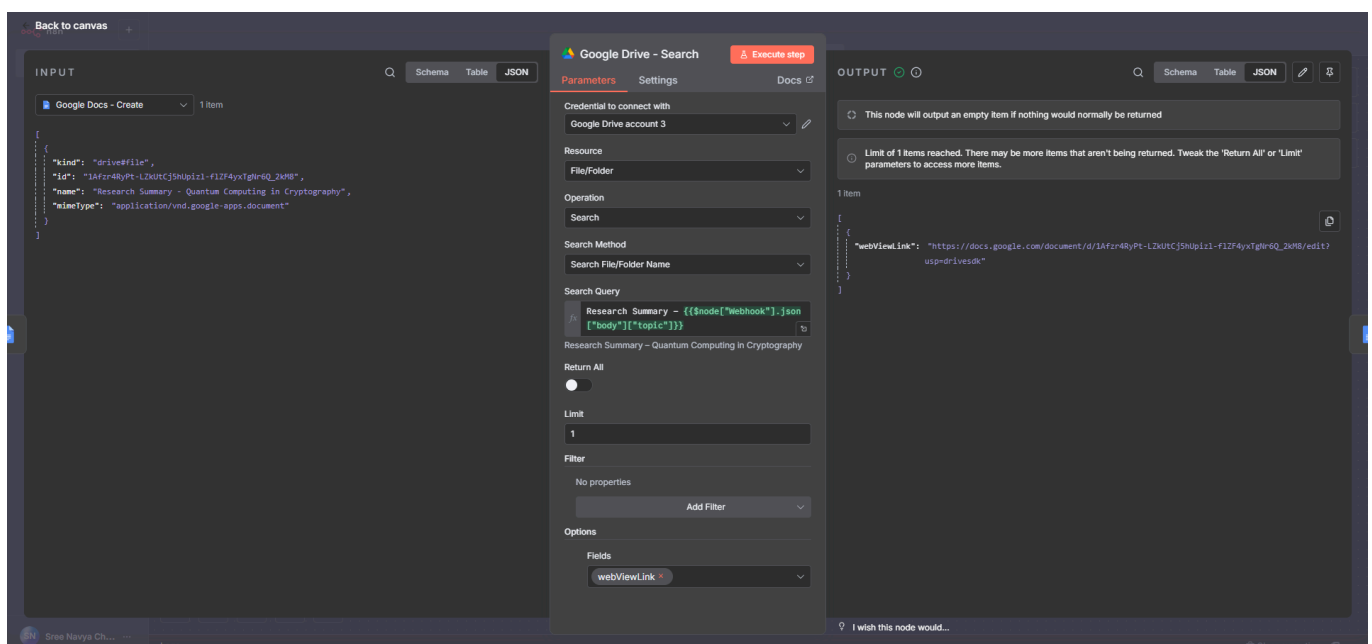
Creates a new Google Document to store the summarized research content.



6 Google Drive Node — “Search Document”

Purpose:

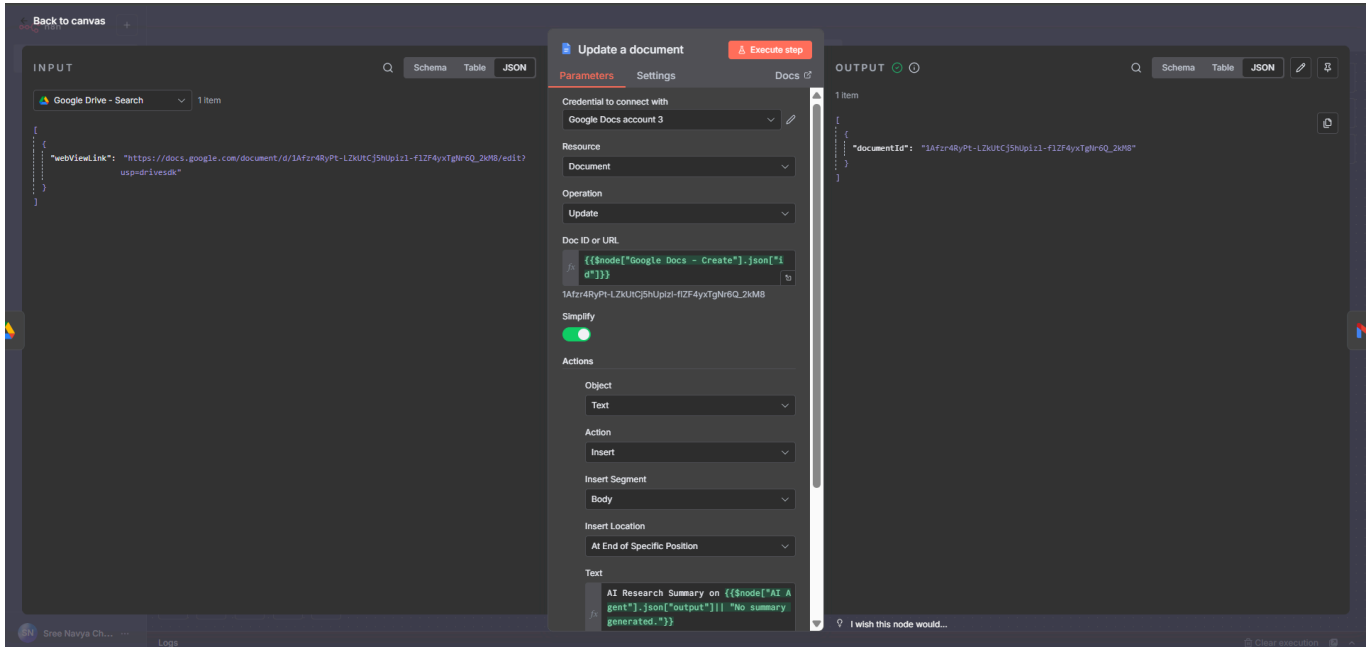
Searches Google Drive to confirm that the created document exists and retrieves its **document ID** for later use.



7 Google Docs Node — “Update Document”

Purpose:

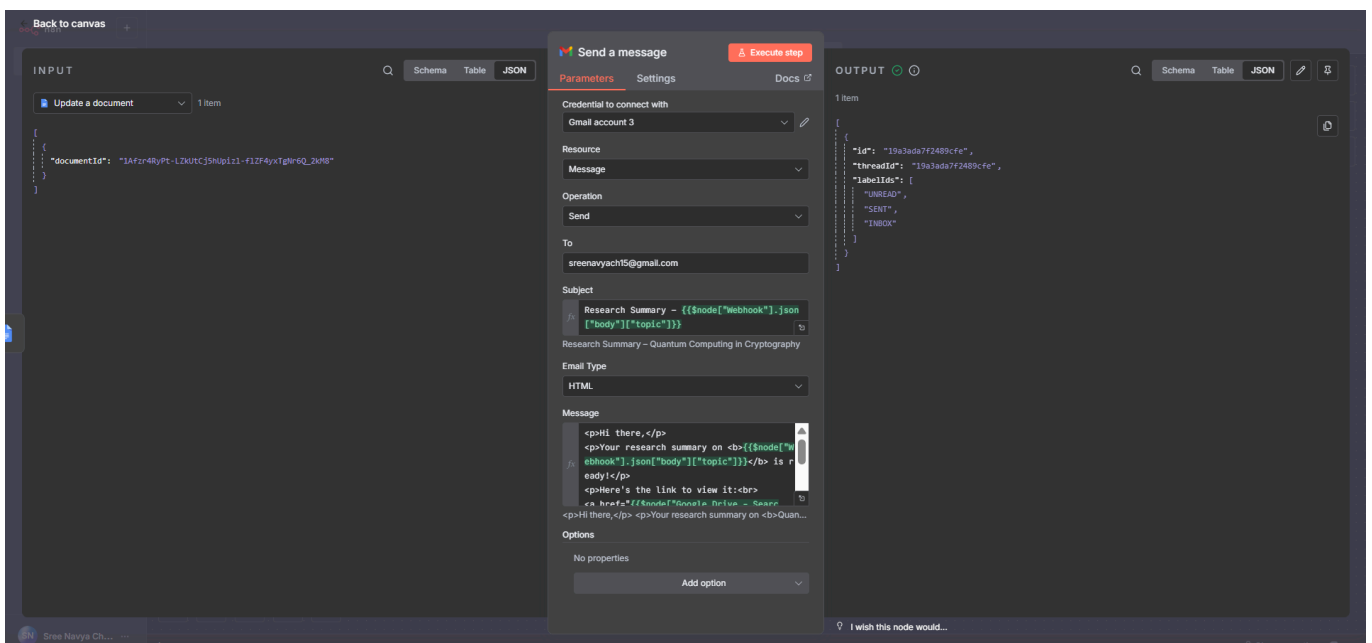
Summarized research contents are stored in the existing document.



8 Gmail Node — “Send a Message”

Purpose:

Automatically sends an email containing the Google Doc as attachment to the user’s Gmail inbox.



Results

The image shows a Gmail email interface at the top and a Google Docs document below it. The email is from 'sreenavyach15@gmail.com' with the subject 'Research Summary – Quantum Computing in Cryptography'. The body of the email says: 'Hi there, Your research summary on Quantum Computing in Cryptography is ready! Here's the link to view it: [Open Google Doc](#). Best, Your Research Assistant'. Below the email is a placeholder for an attachment titled 'Research Summa...'. The Google Docs document is titled 'Research Summary - Quantum Computing in Cryptography' and contains the following text:

AI Research Summary on ## Quantum Computing in Cryptography: A Concise Synthesis

Quantum computing represents a paradigm shift with profound implications across various scientific and technological domains, particularly in cryptography. While still in its nascent stages, the unique capabilities of quantum computers promise both unprecedented security enhancements and existential threats to established cryptographic standards. This synthesis explores the foundational concepts, diverse architectures, and key capabilities of quantum computing as illuminated by current research, framing their direct relevance to the future of cryptography.

At its core, quantum computing leverages quantum mechanical phenomena such as superposition, entanglement, and interference to perform computations. Research into the theoretical underpinnings reveals diverse pathways, from interpreting quantum computation as a geometric evolution process using noncommutative geometry—equivalent to the standard quantum circuit model—to exploring unconventional devices like fermionic quantum computers or those exploiting nonlinear quantum mechanics. These alternative approaches hint at the potential for even faster computation than "conventional" quantum systems, underscoring a rapidly evolving landscape of computational power. A critical aspect of this evolution is the ongoing effort to unify quantum and classical computing, with frameworks like qACP (a quantum generalization of process algebra ACP) seeking to model mixed systems, crucial for verifying complex quantum communication protocols that blend classical and quantum elements.

One of the most significant contributions of quantum computing to cryptography lies in its ability to generate truly genuine random numbers. Unlike classical computers, which are

- Reduced manual effort in literature review.
- Fast and AI-powered research summarization.
- Structured storage and automatic sharing via Google Workspace.
- Demonstrates real-world automation using AI + APIs + n8n.